
Laboratory Guide

Lesson 07 — Topologies and Resilience in ZigBee Networks

Communication Technologies for the Internet of Things

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Estimated duration: 2 hours

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Context

The lecture presented the three device roles and the three topologies admitted by ZigBee, and stated that the mesh topology is more robust because it offers redundant paths. This session verifies that claim: a model of the network is built and the effect of individual node failures is observed.

Physically assembling a ZigBee network requires a coordinator and several modules, which are not available in sufficient numbers. The laboratory ZigBee equipment, the CC2531 dongle, is used in Lesson 13. The present session works on a model, which in fact allows failure scenarios to be tested that would be destructive on real equipment.

No equipment is used and no code is written: the single block provided is only to be pasted.

Material

Personal computer with Node-RED and the `@flowfuse/node-red-dashboard` package.

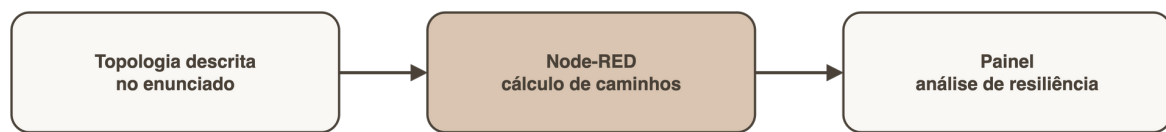


Figure 1: Architecture of the exercise

Learning Outcomes

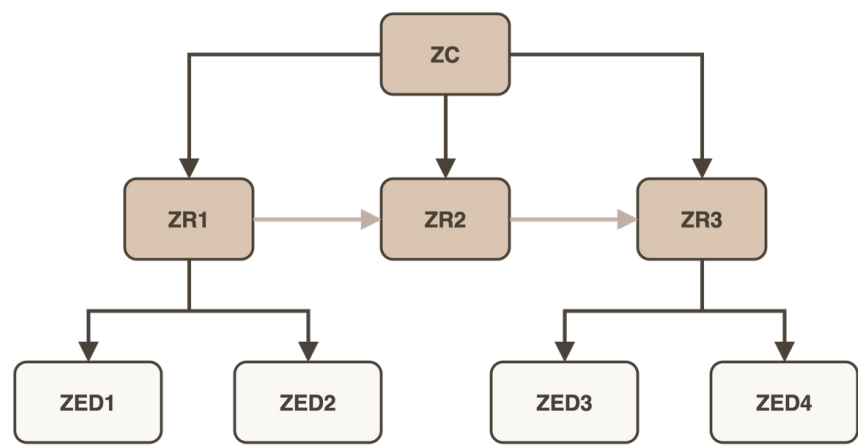
- Represent a concrete ZigBee network in terms of device roles and links;
- Determine, for a given topology, which nodes become isolated on the failure of an intermediate node;
- Compare quantitatively the resilience of the star, tree and mesh topologies;
- Justify the choice of a topology from the requirements of an application.

1. Characterising the Network

25 min

Objective

Describe, in formal terms, the network shown, identifying roles and links.



A tracejado, as ligações redundantes entre routers

Figure 2: Network to be analysed

1. Identify, for each node in the figure, its role: coordinator, router or end device.
2. State which nodes may sleep and which must remain permanently active, justifying from the role assigned.
3. Classify each link as hierarchical, that is, between parent and child, or redundant, between routers.
4. Determine the number of hops between ZED1 and ZED4 along the shortest path.

Node	Role	FFD/RFD	Power and justification
ZC			
ZR1			
ZR2			
ZR3			
ZED1–4			

2. Node-RED Model

40 min

Objective

Build a flow that, given the failure of a node, determines which nodes remain reachable from the coordinator.



Figure 3: Node-RED flow to be built

1. Drag a **ui-dropdown** node from the *dashboard* section, with the list of nodes that may fail: ZR1, ZR2, ZR3 and nenhum.
2. Drag a **function** node with two outputs and paste the provided content:

```

var falha = msg.payload;
var lig = {
  ZC: ["ZR1", "ZR2", "ZR3"],
  ZR1: ["ZC", "ZR2", "ZED1", "ZED2"],
  ZR2: ["ZC", "ZR1", "ZR3"],
  ZR3: ["ZC", "ZR2", "ZED3", "ZED4"],
  ZED1: ["ZR1"], ZED2: ["ZR1"], ZED3: ["ZR3"], ZED4: ["ZR3"]
};
var vistos = {}, fila = ["ZC"];
while (fila.length) {
  var n = fila.shift();
  if (n === falha || vistos[n]) { continue; }
  vistos[n] = true;
  (lig[n] || []).forEach(function (v) { fila.push(v); });
}
var todos = Object.keys(lig);
var ok = todos.filter(function (n) { return vistos[n]; });
var ko = todos.filter(function (n) { return !vistos[n] && n !== falha; });
return [ { payload: ok.join(", ") }, { payload: ko.join(", ") || "nenhum" } ];
  
```

3. Connect the first output to a **ui-text** node labelled *Reachable* and the second to a **ui-text** node labelled *Isolated*.
4. Press **Deploy** and open the panel.

On the provided block

The code traverses the network from the coordinator, skipping the node declared as failed, and returns the nodes reached and those left unreachable. It is provided in full and is not assessed; what is assessed is the reading of the results.

3. Failure Trials

35 min

Objective

Determine experimentally the effect of each router failing and relate the result to the topology.

1. For each node in the list, record the nodes that become isolated.
2. Identify which router failure has the greatest impact and explain why from the figure.
3. Remove from the `lig` object the redundant links between routers, turning the network into a tree, and repeat the trial. Record the difference.

Failed node	Isolated (mesh)	Isolated (tree)
ZR1		
ZR2		
ZR3		

Expected result

The failure of ZR2 has different effects in the two topologies. In the mesh, traffic bypasses it through the direct links to the coordinator; in the tree, the entire branch depending on it becomes unreachable. It is this difference that supports the claim, made in the lecture, that the mesh is more robust.

Further work

Add to the `lig` object a fourth router and two additional end devices, and determine how many redundant links are needed so that no single failure isolates any node.

Questions for Discussion

1. Why do end devices never appear as points of failure for other nodes?
2. The mesh topology has costs. Which, compared with the star topology?
3. In a home automation deployment with thirty devices spread over three floors, which topology would you choose, and why?
4. Would the frequency agility mechanism mentioned in the lecture resolve any of the failures tested? Justify.

References

- Gonçalves, P. *Apresent6 — Zigbee*, TCIoT, previous editions.
- Ramya, C. M. *et al.* (2015). A Survey of ZigBee Wireless Sensor Network Technology: Topology, Applications and Challenges. *IJCA* 130(9).
- Node-RED documentation: <https://nodered.org/docs/>